

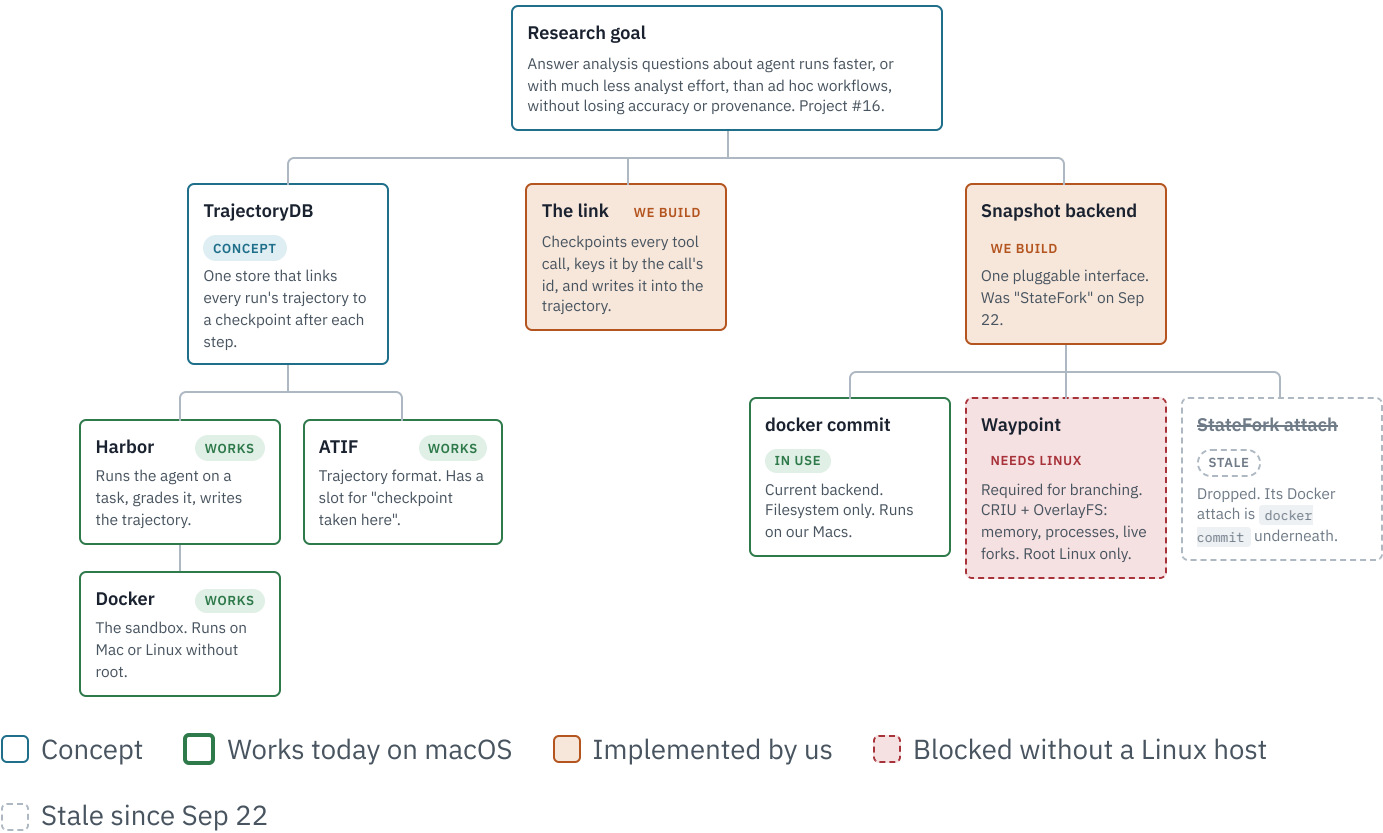
Branching needs Linux; comparison needs TB 2.1

TWO DECISIONS ON FRIDAY

1. The project brief requires branchable environment state. Only Waypoint branches a live sandbox, and it runs only on a root Linux host, so we need a Linux server. 2. Comparison is the core of our contribution, and a 52-source survey shows the field compares on Terminal-Bench 2.0/2.1 and never on 3.0/4.0, so the corpus should target TB 2.1, not the live 4.0 set.

Revised architecture

This is the same tree as Sep 22 with the state half corrected. StateFork is the common interface and benchmark layer. The CRIU engine that captures memory and forks live sandboxes is Waypoint. StateFork's Docker path is `docker commit`, the same thing we already run, so we dropped it (ADR-0004).



Three blockers

We checked this against the StateFork and Waypoint source (v0.7.0, vendored under `vendor/`) and the Harbor source. A Linux server removes the first blocker. The second is integration work we can do without patching Harbor. The third rules out every shortcut.

- **It needs root on a Linux host.** Waypoint is a Go binary built on CRIU and OverlayFS, and it requires root. On macOS, Docker Desktop's VM has no CRIU at all (docker/for-mac#1059, open since 2017). OrbStack exposes `docker checkpoint`, but only for arm64-native containers, and it is still not a Linux host we control.
- **It does not attach to Harbor's containers.** Harbor starts every task with `docker compose`. Waypoint stages its own rootfs and PTY sessions, and `waypoint_attach` takes a Waypoint session id, not a container or a PID. The way in is a custom Harbor environment, loaded with `--env module:Class`, that builds each task with `waypoint build` from the task's Dockerfile and runs commands with `waypoint exec`. Harbor stays unpatched. Open risk: tasks that need several compose services or network isolation.
- **The other CRIU routes are dead ends.** StateFork's `docker_attach` is `docker commit` (`container_env_manager.py:120`). `hybrid_attach` captures memory, but only on Podman with runc and root, so Harbor runs would have to move off Docker. Docker's own `docker checkpoint` is still experimental and fails on containers with an external terminal, which is how agents run.

CRIU is also expensive at our granularity. Each dump is roughly as large as the process memory (about 1.7 s per 2 GB, arXiv:2510.05556), and we would take one for every state-changing tool call in every trial.

What Harbor and Docker still record

RECORDED ON A MAC TODAY

- The full ATIF trajectory for every trial (Harbor)
- Grade, tokens, and cost per trial
- An environment checkpoint after every tool call: hook, then watcher, then `docker commit`
- The join: each checkpoint keyed by `tool_call_id` and inserted into the trajectory
- Each checkpoint is a runnable image, so `docker run` shows the filesystem at step N

LOST WITHOUT WAYPOINT

- Memory and running processes (servers, daemons)
- Shell state: working directory, exported variables, background jobs
- Native live branching: forking a running sandbox mid-trial into concurrent copies
- Exact restore: a commit image restarts cold, with a fresh shell and no processes

What dropping Waypoint costs

DIMENSION	DOCKER COMMIT (NOW)	WAYPOINT (CRIU)	COST TO US
Filesystem state	Captured	Captured	None
Memory, processes, shell	Lost	Captured	Low. Terminal-Bench grades files, and our analysis questions are framed around filesystem changes. Most shell side effects that matter end up on disk.
Branching	Cold only: start a new container from the step-N image	Concurrent live forks from any checkpoint	Crucial. The project brief asks for branchable environment state. <code>docker commit</code> alone does not meet it.
Host	Mac or Linux without root	Root Linux only	Blocks our Macs. None on a server.
Harbor integration	Attaches to its containers as they are	Needs a custom Harbor environment	Medium. Integration work. Harbor stays unpatched.
Storage per checkpoint	Re-tars the writable layer; commits do not dedupe	Memory-sized dump, about 1.7 s per 2 GB	Low for Terminal-Bench-sized write sets

Dropping Waypoint fails the brief. ADR-0004 needs to be reopened. `SnapshotBackend` is pluggable, so the capture pipeline carries over, with a new `corpus_id`.

Terminal-Bench on Apple Silicon

This is a second, independent reason to get a Linux server. Harbor does not pin a platform, so on our M-series Macs the containers run as arm64. Some Terminal-Bench tasks are x86-only. Our Sep 24 static scan of the full registry (every Dockerfile, image manifest, and x86 payload) quantified it: 5 of the 66 tasks on current main are arm64-broken, about 8%. The sixth task below existed only in the v3.0.0 registry and is removed on main.

TASK	ISSUE	FAILURE ON ARM64
fix-uautomizer-soundness	#1601	Silent: all verdicts UNKNOWN, even the reference solution scores 0. Looks like an ordinary agent failure. v3.0.0 only; removed on main.

TASK	ISSUE	FAILURE ON ARM64
coq-block-bound	#1802	Image not multi-platform
hof-topology-interpenetration	#1803	Image not multi-platform
math-eval-grader	#1804	Image not multi-platform
gsea-proteomics	#1805	Image not multi-platform
wdm-design	#1806	Image not multi-platform

- **The list is version-scoped.** The scan covers the 3.0/4.0 registry only. TB 2.1's 89 tasks have never been arm64-scanned, so if we adopt 2.1 this exclusion list does not transfer and Mac development runs need a fresh scan. The recorded corpus runs on x86_64 Linux either way, where none of this applies.
- **Silent failures can hide.** Cross-platform CI is only proposed (#1793). Because #1601 fails silently, other broken tasks can hide in a Mac-recorded corpus as fake agent failures, and we cannot measure that bias.
- **Emulation is for development only.** `DOCKER_DEFAULT_PLATFORM=linux/amd64 harbor run ...` runs x86 images under emulation, and the #1601 reference solution then scores 1. Emulation distorts timing (checkpoint latency, task timeouts), so emulated runs cannot go into a recorded corpus.

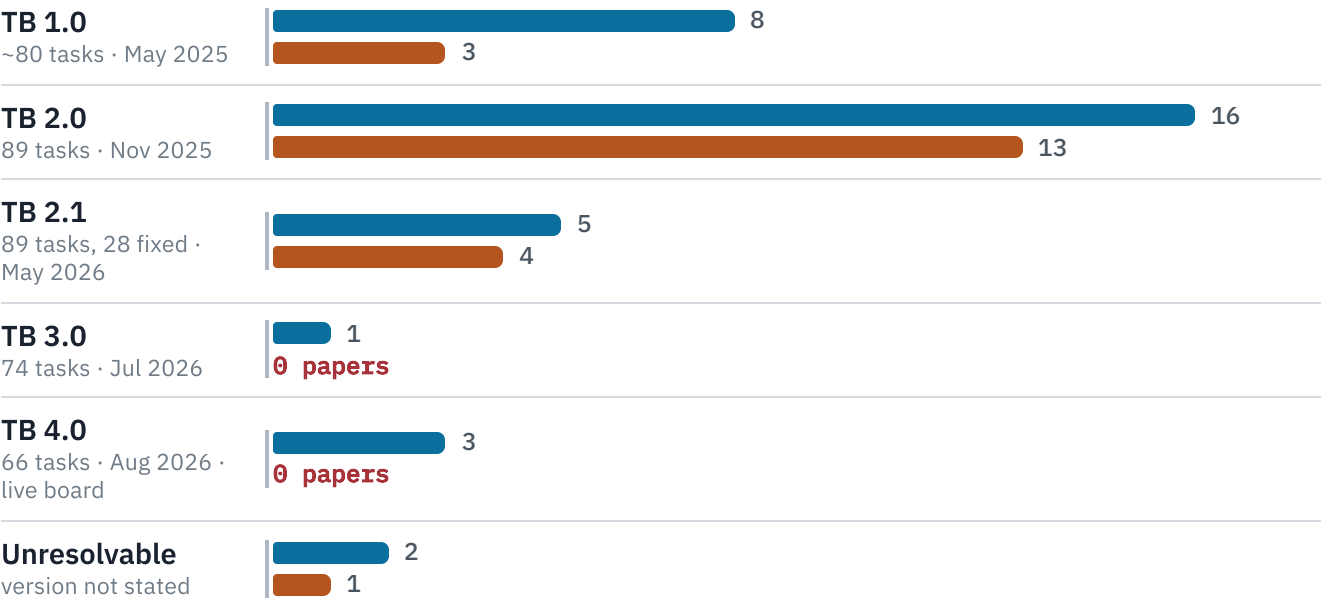
Which Terminal-Bench version: the Sep 24 survey

Comparison is the core of our research contribution, so the corpus task set should be the version the field actually compares on. On Sep 24 we surveyed 52 primary sources - 34 model reports and system cards, 18 research papers - and verified every version claim against the primary document (arXiv full texts, system cards, HF cards, leaderboard snapshots). Full evidence tables: `docs/research/2026-09-24_tb-version-survey.md`.

CORRECTION TB 2.1 exists. Our Sep 24 morning note said it had no released artifact; it shipped 2026-05-06 as Harbor Hub dataset `terminal-bench/terminal-bench-2-1` (no git tag, which is why a tag-only search missed it). It is TB 2.0's 89 tasks with 28 broken tasks repaired, and it has its own frozen leaderboard.

Who runs what (56 version calls across the 52 sources)

Model reports and cards (34) Research papers (18)



Four sources run two versions, so calls sum to 56. Zero academic papers have ever evaluated on 3.0 or 4.0; TB 4.0 appears only in the three newest lab reports (all Sep 2026) and on the live leaderboard. The 89-task 2.0/2.1 family carries 38 of the 56 calls, and the newest RL and agent papers (T1, StateM, Tmax) plus the mid-2026 frontier claims (Claude Fable 5 at 84.3%, Grok 4.5, GLM-5.3) are on 2.1.

The two real contenders

TB 2.1 - RECOMMENDED

- The 28 repaired tasks remove a known source of spurious failure trajectories in the corpus
- Where recent papers and frontier claims live; its frozen leaderboard (top entries 78-83%) stays a stable anchor
- GLM-5.3 already ran it under the Claude Code harness - exact precedent for our setup
- Per-task timeouts (900 s / 1800 s tiers) keep docker-commit checkpoint chains bounded
- Concern: near saturation (~84% frontier), so only ~15-20% failure trajectories, mostly in the 30-task hard tier

TB 4.0 - LIVE LEADERBOARD

- Zero academic comparables; only three Sep 2026 lab reports
- SOTA ~58%, so better headroom if our claim ever becomes "our method improves the agent"
- Flat 8-hour agent timeout and GPU-requiring tasks multiply run cost and checkpoint storage
- Leaderboard numbers are judge-filtered since Apr 2026, so self-run numbers are not exactly board-comparable either

Comparability rules that hold whichever we pick: harness choice alone is worth 6-10 points (T1 measured Opus 4.6 at 70.1 under Claude Code vs 63.8 under Terminus-2), 2.1 scores run ~10 points above 2.0-era numbers because of the repairs, and the leaderboard protocol is k=5 mean while Harbor's default is `n_attempts=1`. Any number we publish must pin harness, attempt count, and aggregation.

TEAM DECISION FOR FRIDAY

Adopt TB 2.1 as the primary corpus version? Recommendation only so far - the repo rule requires an ADR and a new corpus_id before the first run. Open items before the ADR:

- Read the Claude Fable 5 system card PDF and confirm the harness behind the 84.3% claim (currently secondary-sourced)
- Pin the Hub dataset `terminal-bench/terminal-bench-2-1` version against our pinned Harbor and record it in the job config
- Choose the attempt protocol: k=5 mean (leaderboard) vs avg@3 (GLM-5.3's protocol)
- Pilot the hard tier to count expected failure trajectories; keep a TB 4.0 hard-subset secondary corpus (own corpus_id) as fallback if failures come up thin

Request a Linux server on Friday

One Linux machine fixes both problems. Root lets Waypoint branch live sandboxes, which the brief requires. x86_64 removes the architecture bias.

QUESTION FOR THE PROFESSOR

Can the course or department provide or fund an x86_64 Linux server or VM with root access to record the corpus?

- x86_64, with root, running Docker Engine (not Docker Desktop)
- Enough disk for one committed image per tool call in every trial
- Can run long batch jobs unattended
- A kernel that supports CRIU, which Waypoint needs for branching
- Disk for CRIU dumps, which grow with process memory

Development stays on our Macs until the server arrives, with emulation for x86-only tasks. The server blocks branching and the recorded corpus, not the Sep 25 baseline (stock Harbor, no checkpoints). Team decisions for Friday: reopen ADR-0004 and plan the Waypoint environment, and settle the corpus version (recommendation: TB 2.1, pending the open items above).

Sources: docs/project-brief.md, docs/decisions/0004-file-system-only-checkpoints.md, docs/research/2026-09-23-checkpoint-platform-research.md, docs/research/2026-09-24_tb-version-survey.md and 2026-09-24_tb-version-recommendation.md, vendor/statefork and vendor/waypoint at v0.7.0, arXiv:2510.05556, docker/for-mac#1059, harbor-framework/terminal-bench issues #1601, #1793, #1802-#1806, tbench.ai news and leaderboards. Supersedes the StateFork section of the Sep 22 brief. State as of Sep 24, 2026.